

(12) **United States Patent**
Runser

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(54) **PHOTOVOLTAIC MODULES AND ROOFING SHINGLES WITH NAIL ZONES**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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H02S 20/25 (2014.01)

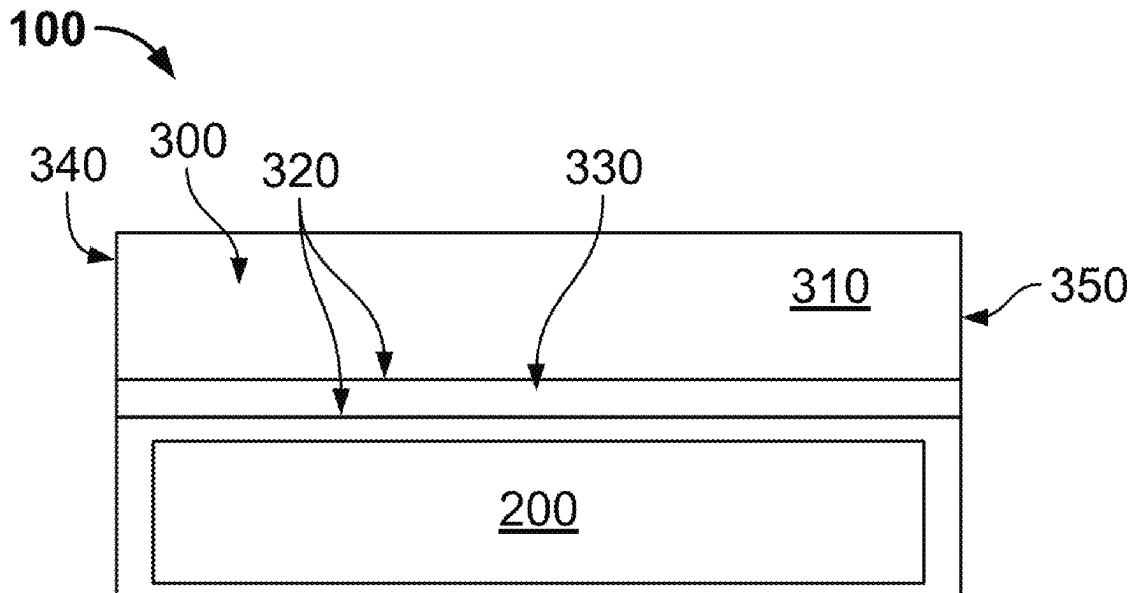
(52) **U.S. Cl.**
CPC **H02S 20/25** (2014.12); **E04D 1/34** (2013.01); **E04D 2001/3423** (2013.01)

(58) **Field of Classification Search**
CPC E04D 1/34; E04D 2001/3423; E04D 2001/3452; E04D 2001/3458; E04D 2001/3467; E04D 2001/3491; H02S 20/25
See application file for complete search history.

ABSTRACT

A system, including: a roof deck; a plurality of photovoltaic modules installed on the roof deck, each of the plurality of photovoltaic modules having: a solar cell, and a backsheet, where the solar cell is positioned on the backsheet, where the backsheet has a headlap portion, where the headlap portion has a first longitudinal groove and a second longitudinal groove, where the first longitudinal groove is spaced apart from the second longitudinal groove on the backsheet, to define a nail zone between the first longitudinal groove and the second longitudinal groove; and a plurality of fasteners installed so as to penetrate the nail zones of the plurality of photovoltaic modules and the roof deck.

20 Claims, 12 Drawing Sheets



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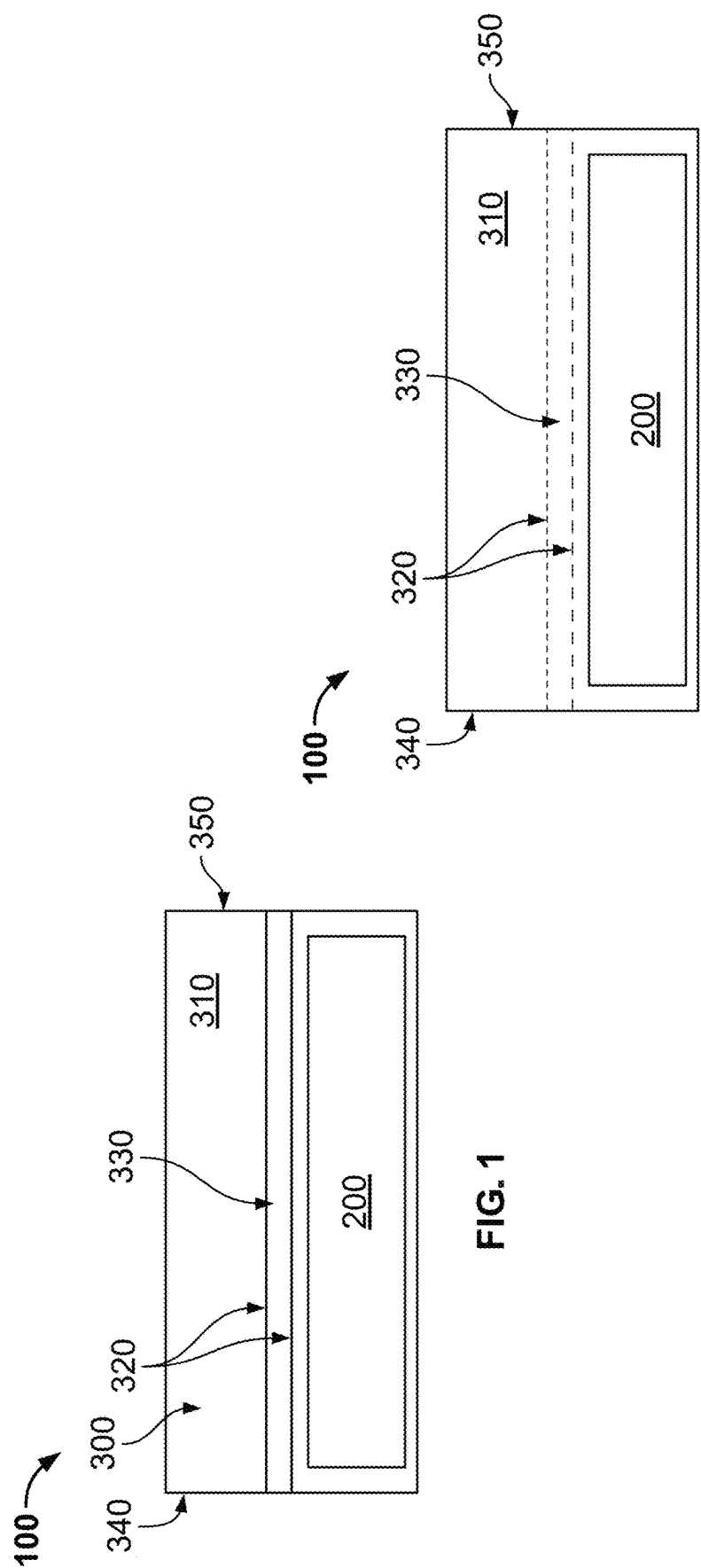


FIG. 1

FIG. 2

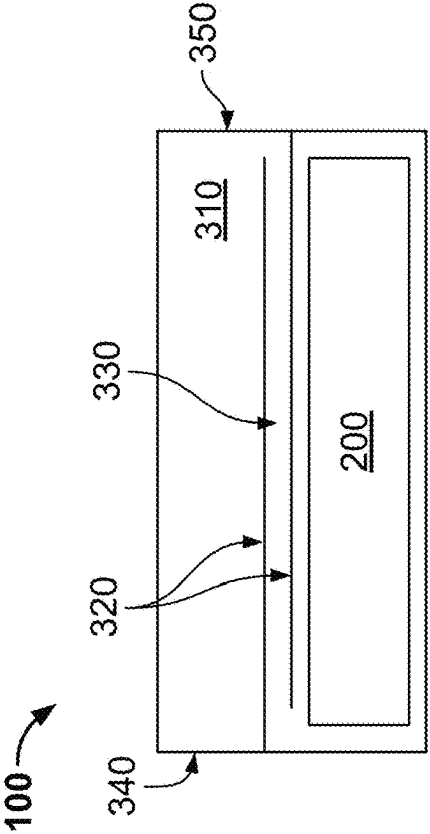


FIG. 3

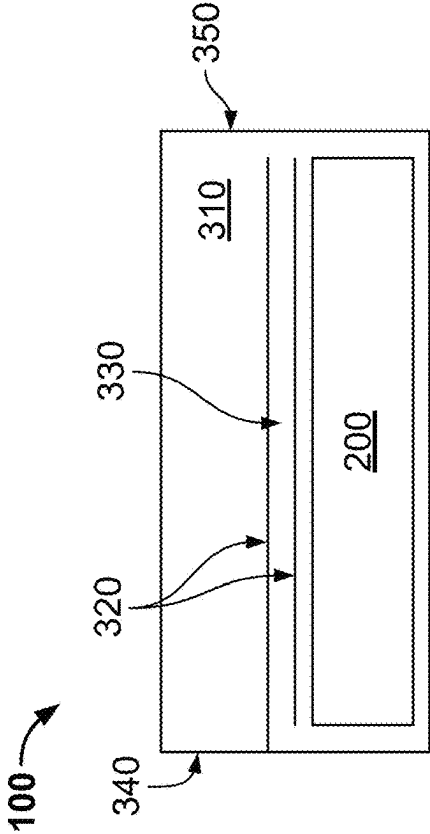


FIG. 4

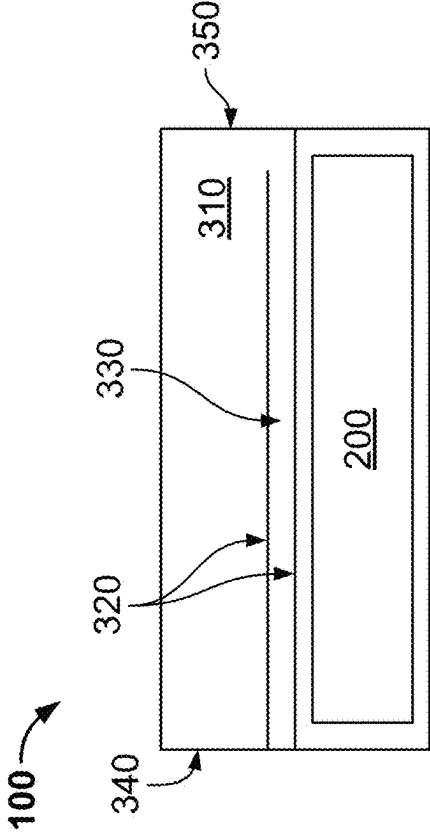


FIG. 5

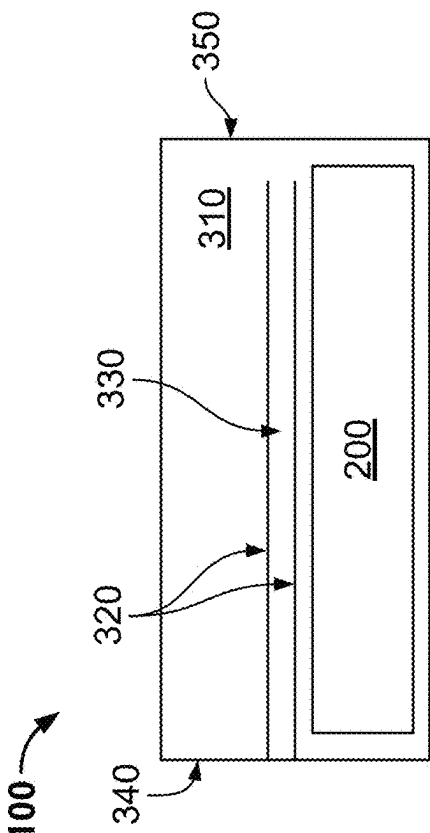


FIG. 6

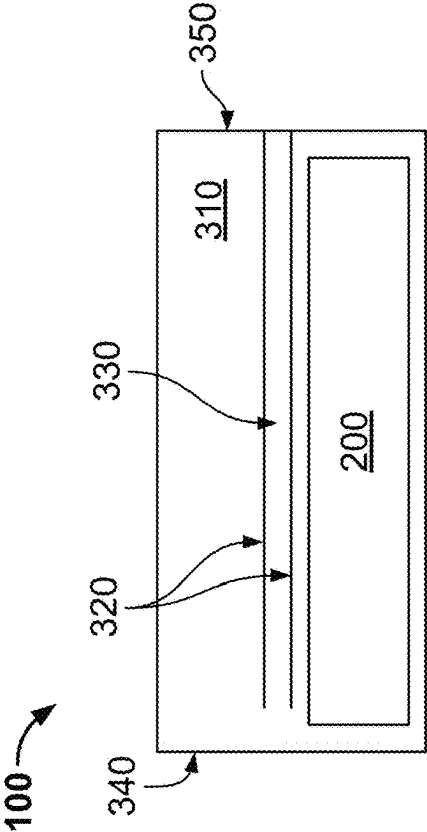


FIG. 8

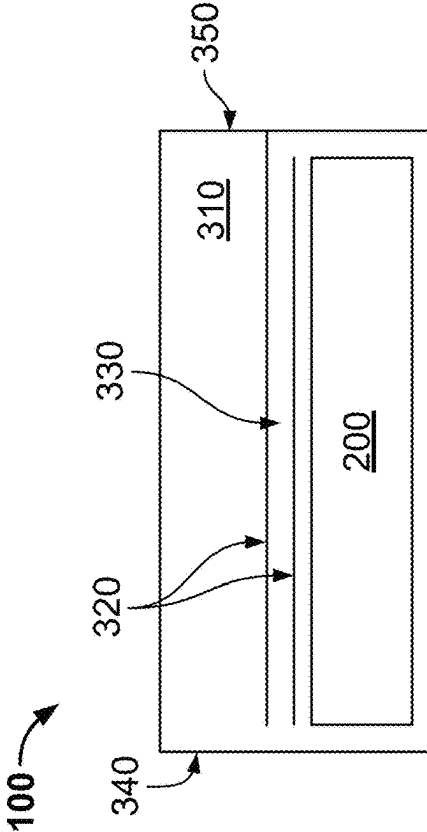


FIG. 10

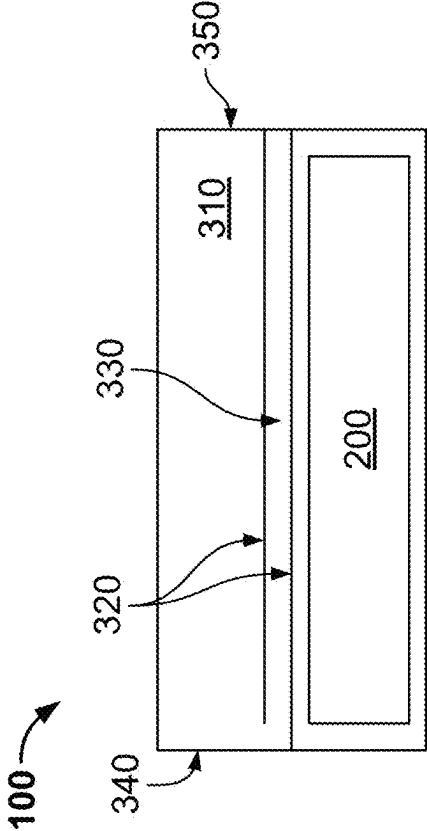


FIG. 7

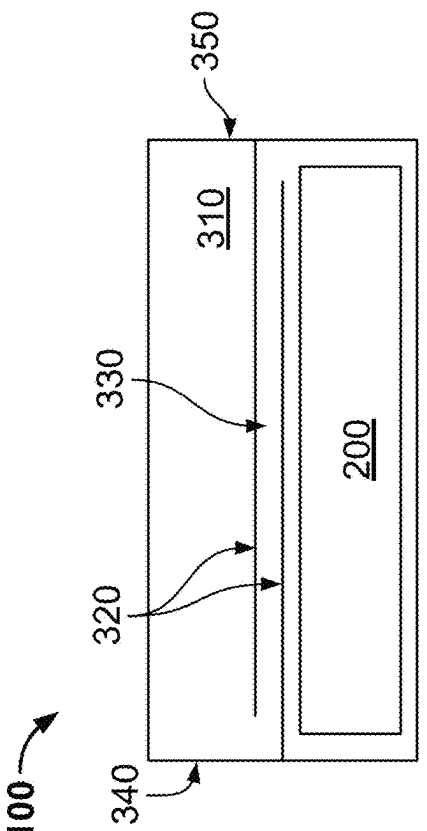


FIG. 9

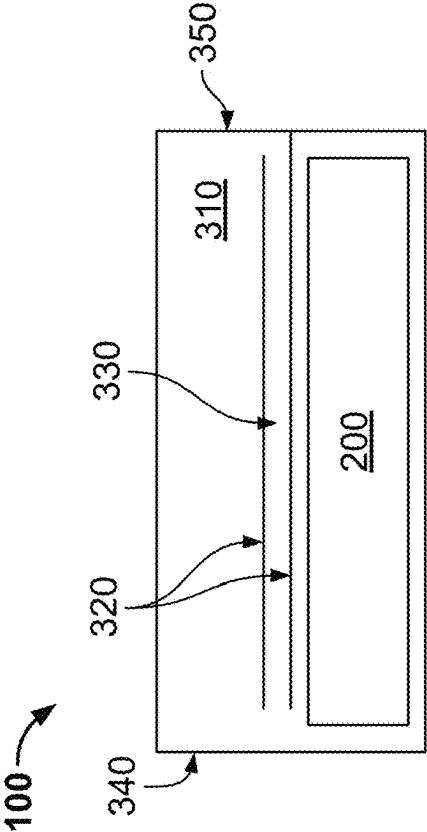


FIG. 11

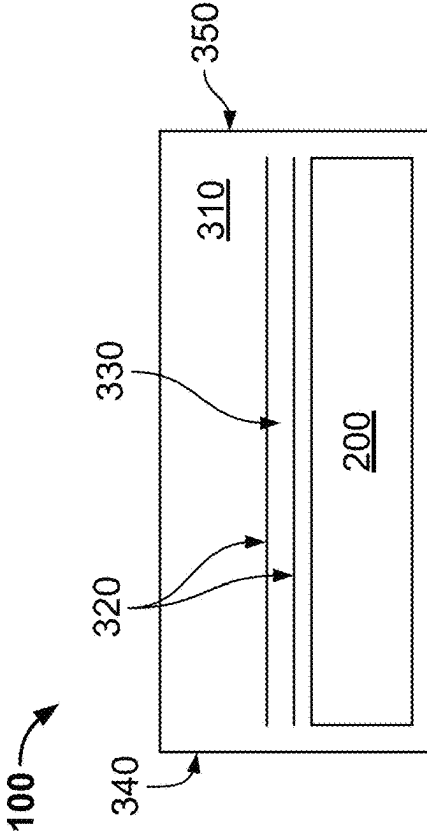


FIG. 12

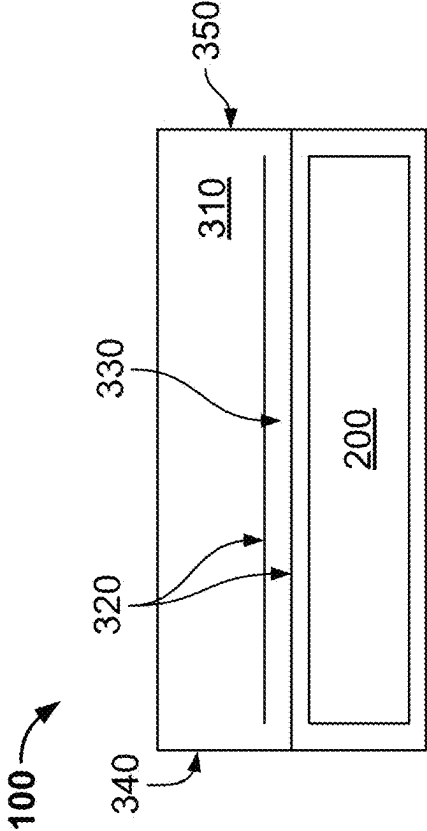


FIG. 13

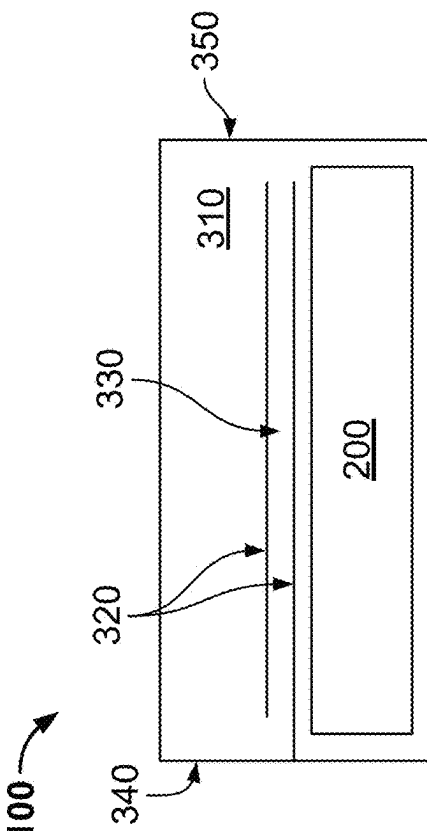


FIG. 14

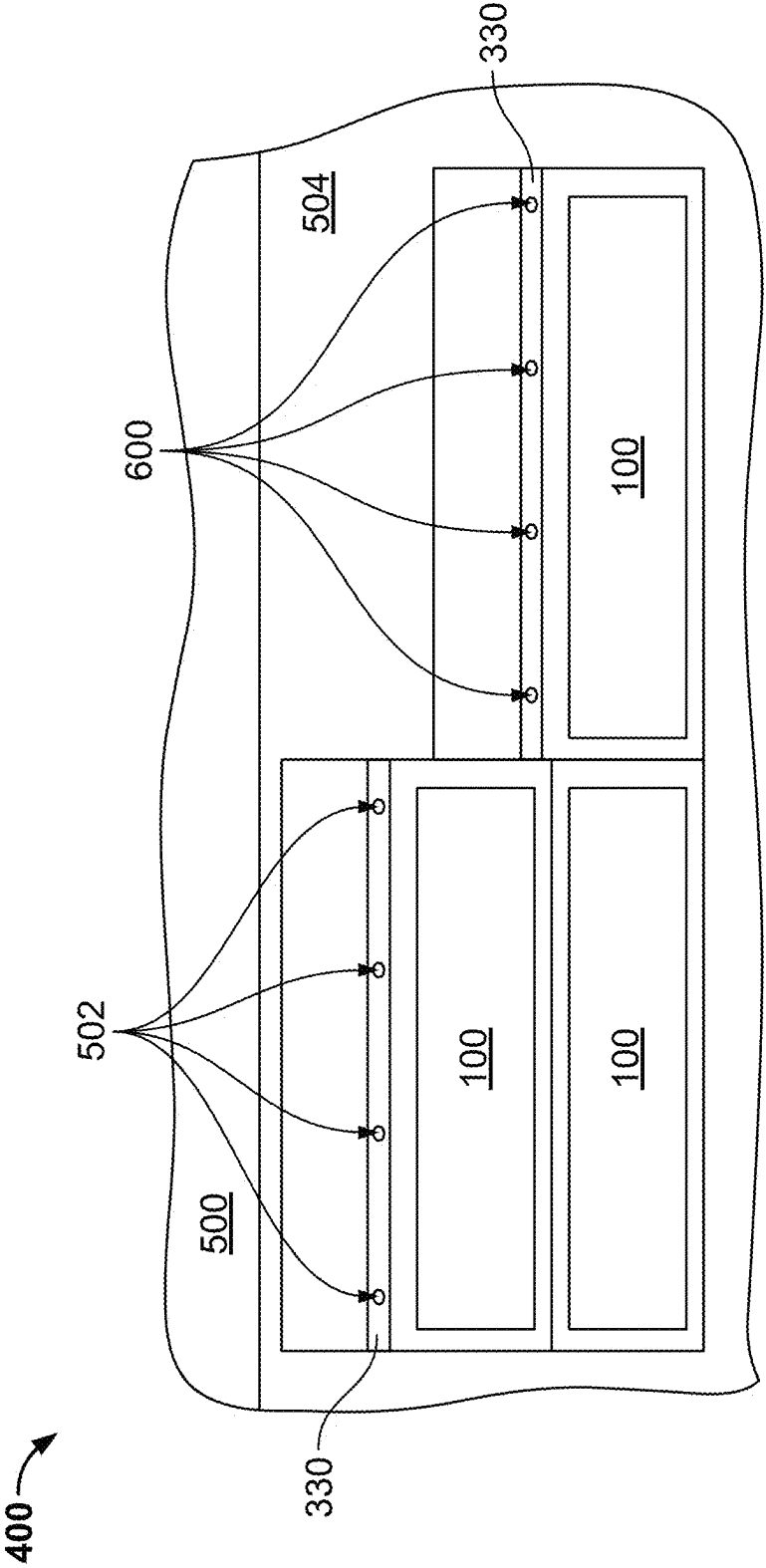


FIG. 15

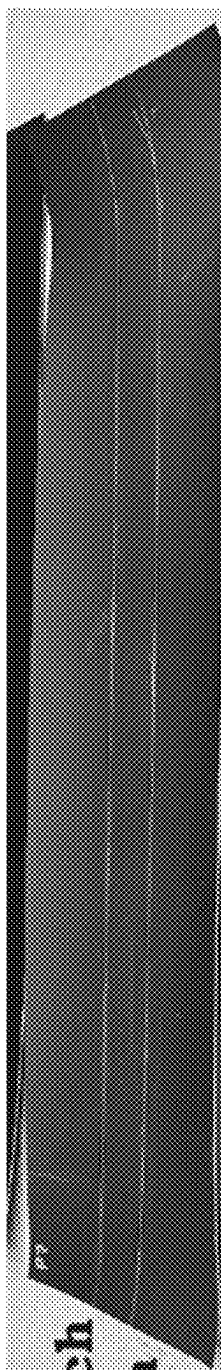


FIG. 16

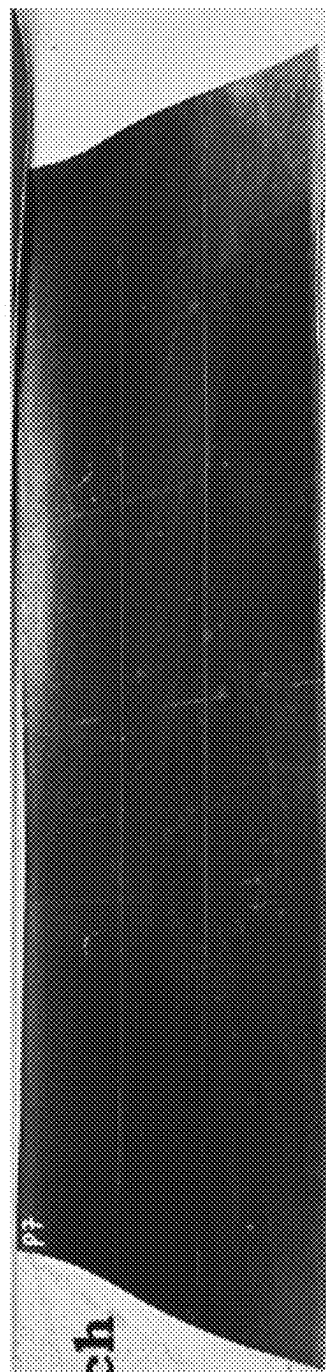
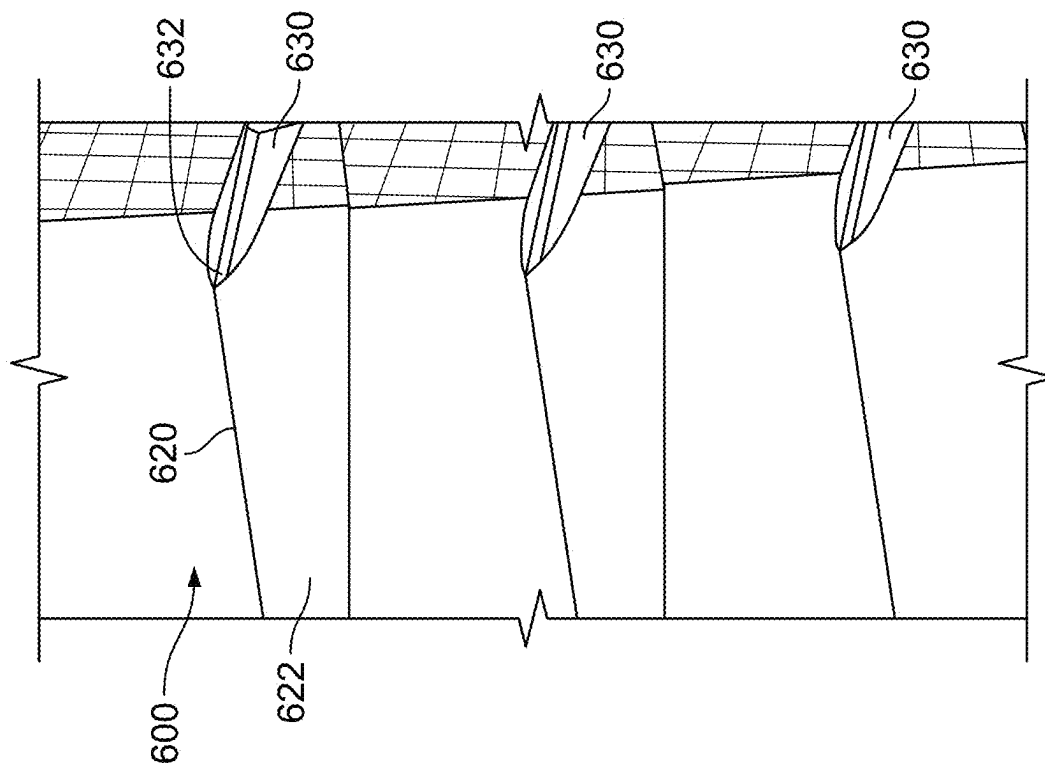


FIG. 17



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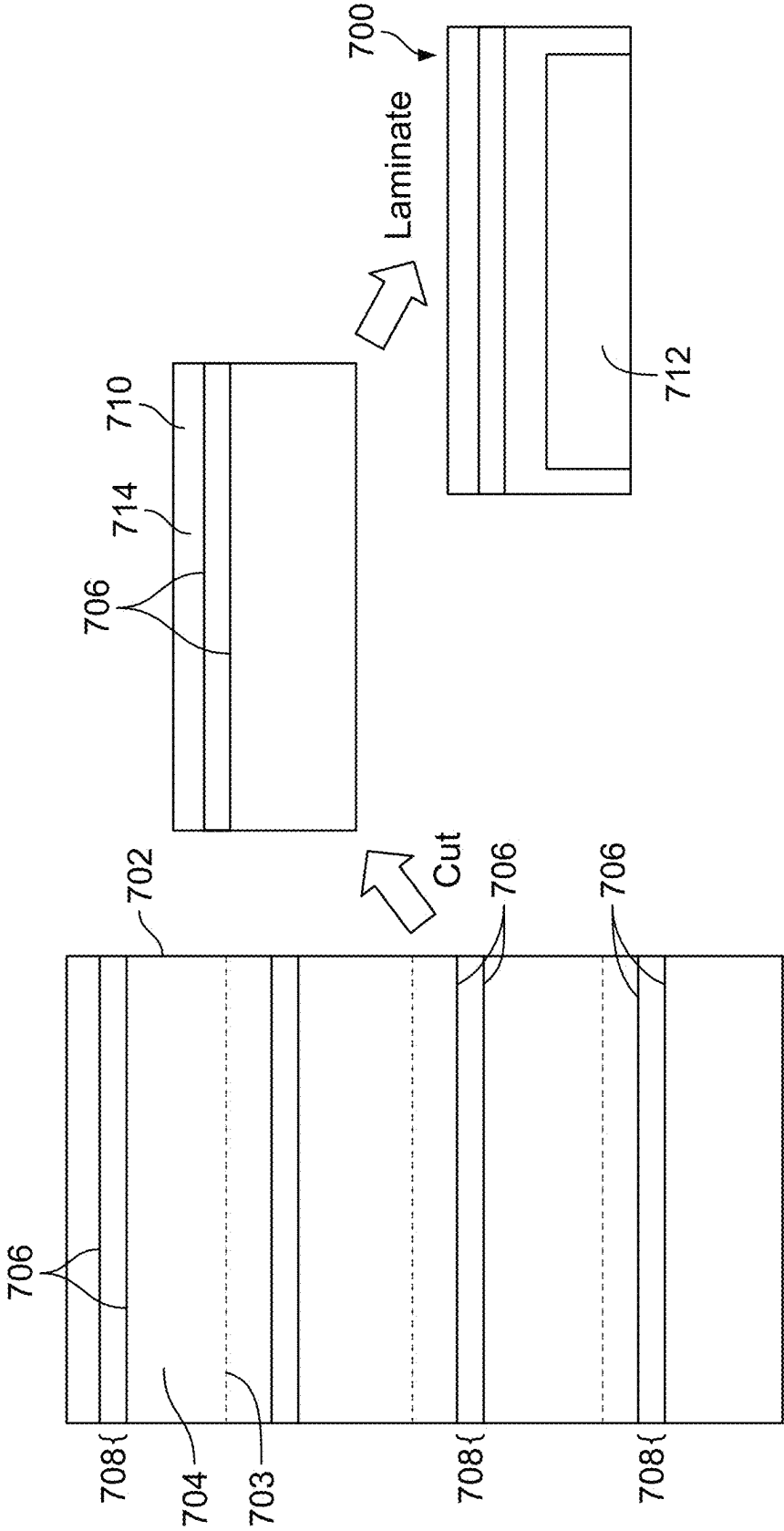


FIG. 19

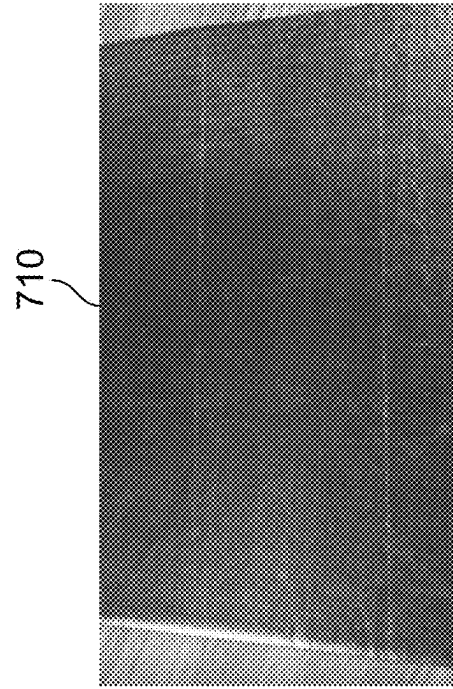


FIG. 20A

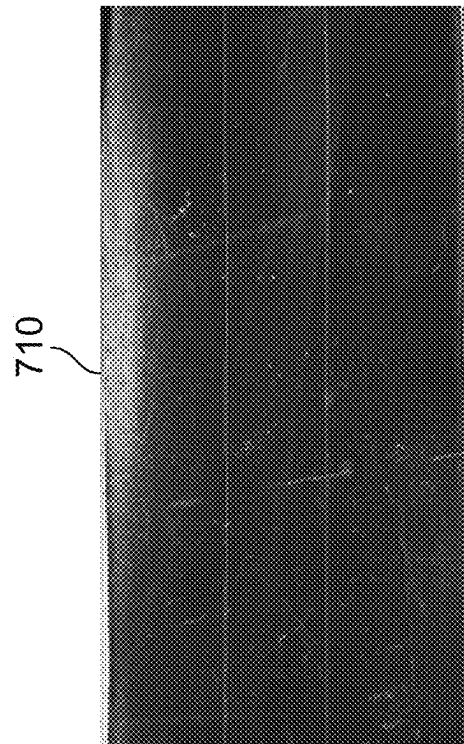


FIG. 20B

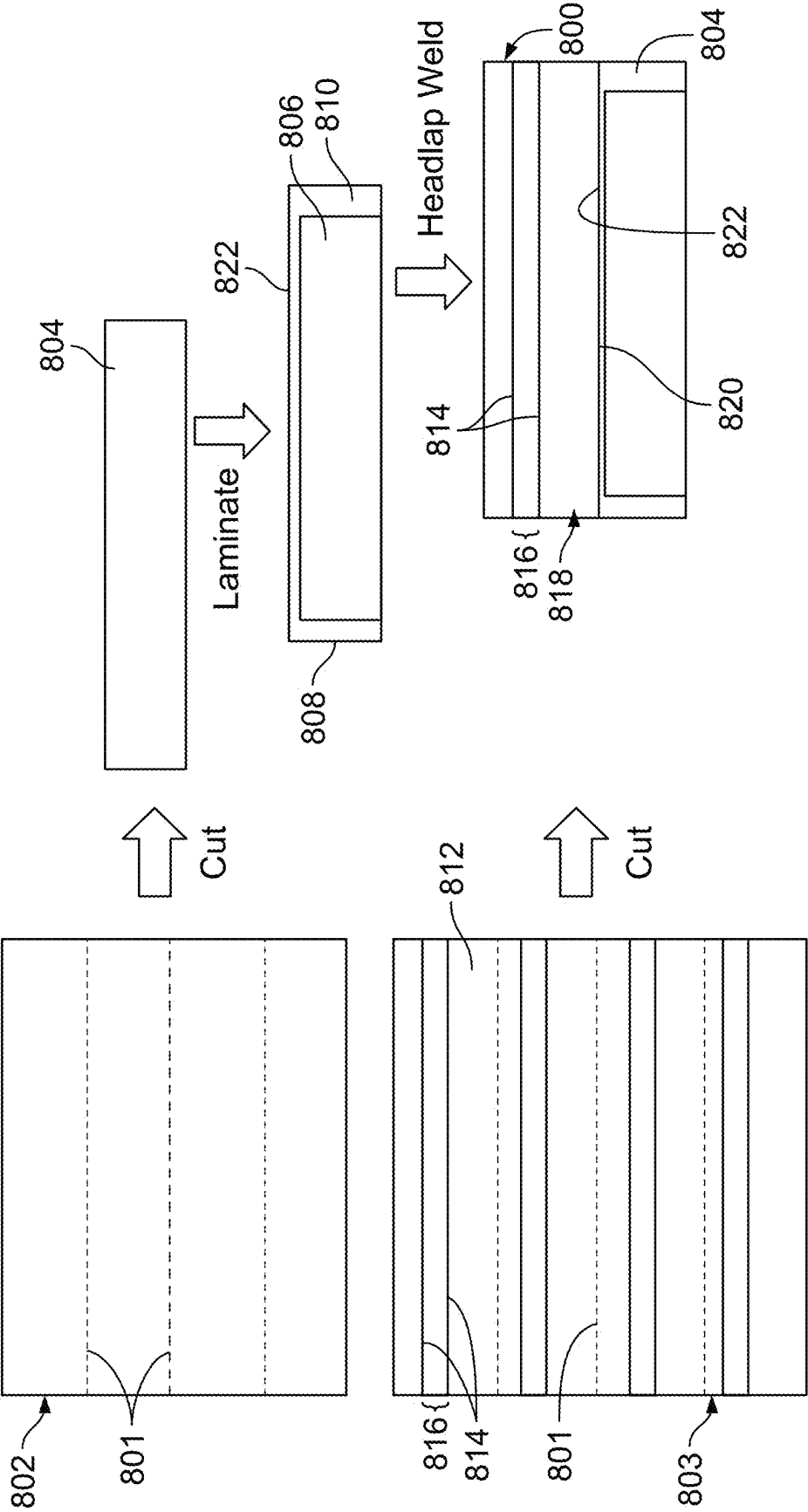


FIG. 21

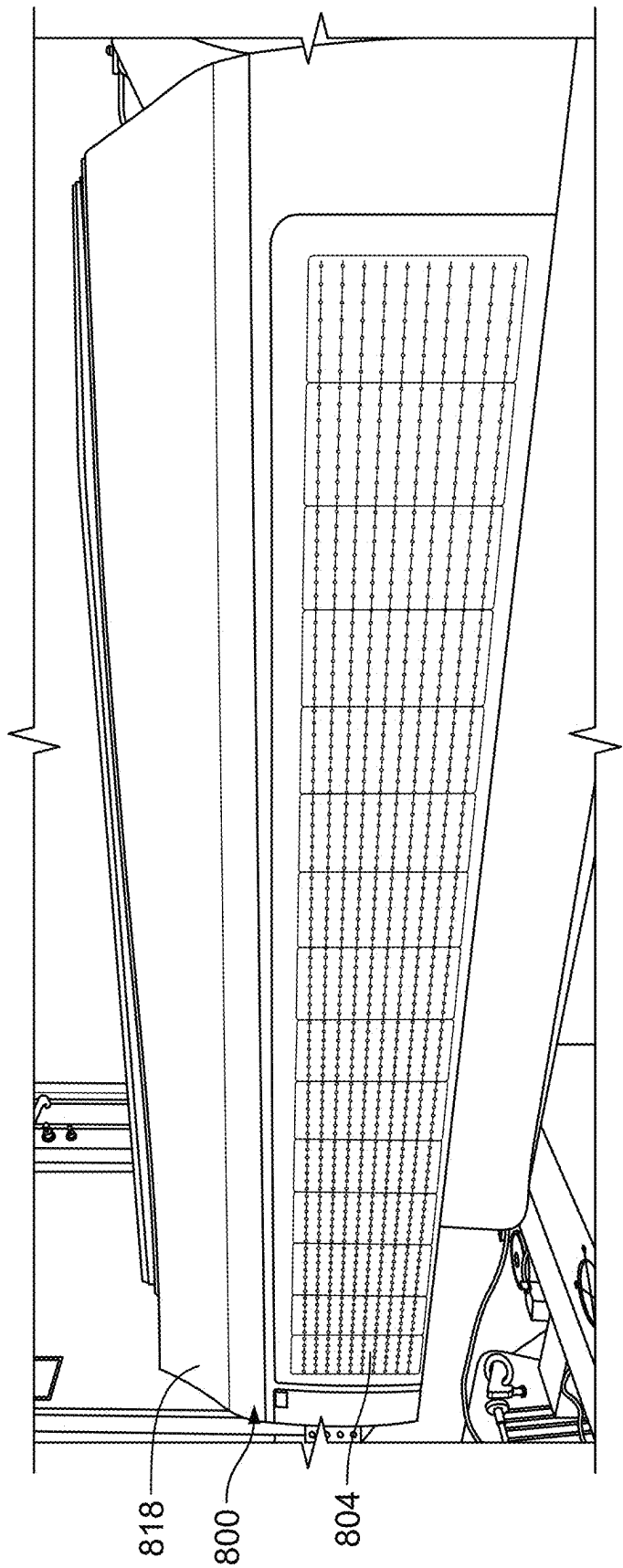


FIG. 22

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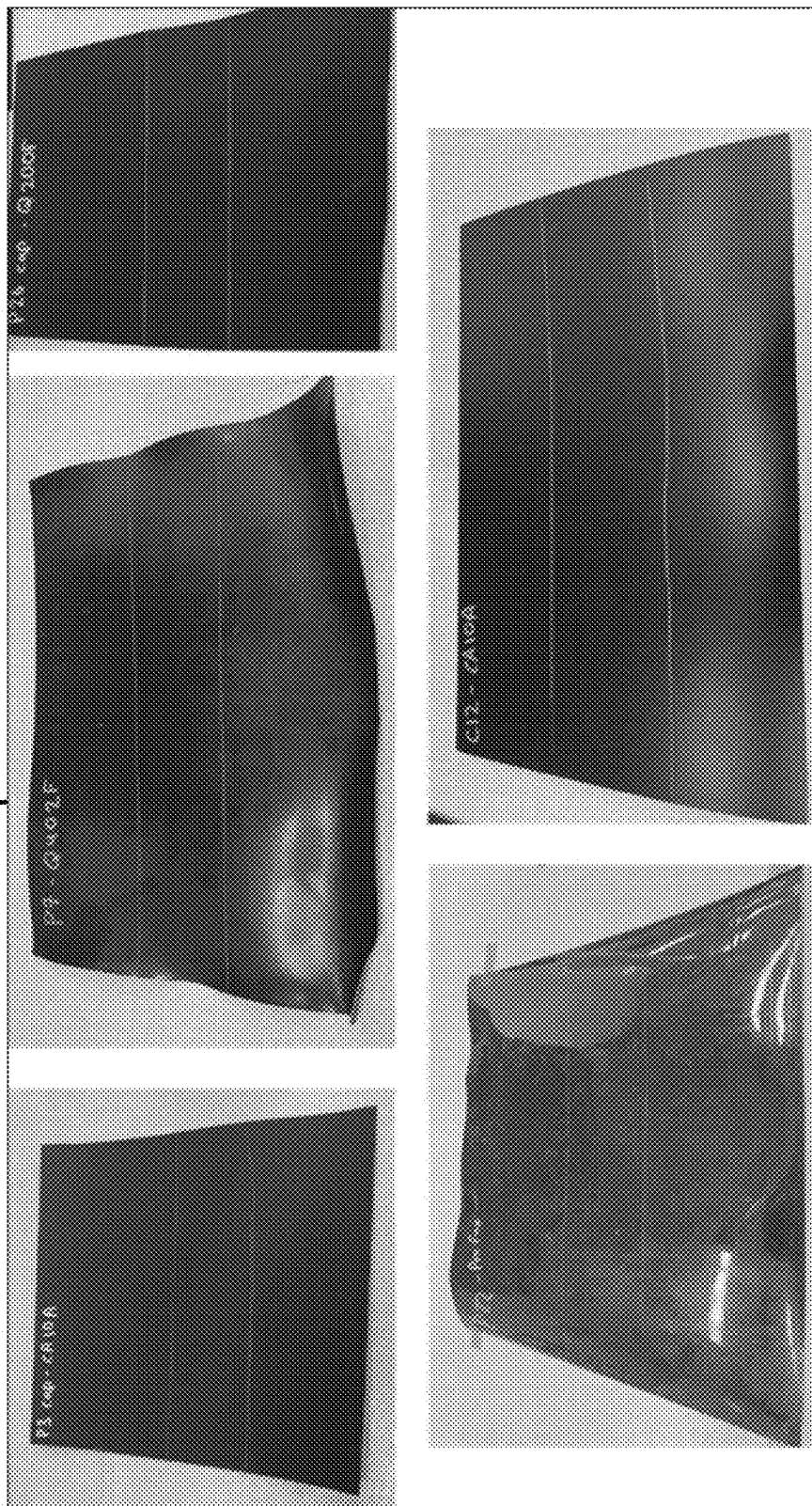


FIG. 23

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PHOTOVOLTAIC MODULES AND ROOFING SHINGLES WITH NAIL ZONES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Section 111(a) application relating to and claiming the benefit of commonly-owned, U.S. Provisional Patent Application Ser. No. 63/579,858, filed Aug. 31, 2023, entitled "ROOFING MODULE, AND ASSOCIATED SYSTEM AND METHOD," the contents of which are incorporated herein by reference in its entirety.

FIELD

The present invention relates to roofing modules including photovoltaic modules and roofing shingles and, more particularly, photovoltaic modules and roofing shingles with nail zones.

BACKGROUND

In a roofing system, a technician may install a number of roofing modules, including both photovoltaic modules and non-photovoltaic modules, above a roof deck. For example, the technician may fasten a number of fasteners through a nail zone of each of the roofing modules, and into the roof deck.

SUMMARY

The Claims, rather than the Summary, define covered embodiments of the present invention. The Summary is a high-level overview of various aspects of the invention, and introduces some concepts that are further described in the Detailed Description below. The Summary is not intended to identify key or essential features of the claimed subject matter, and also is not intended to be used in isolation to determine the scope of the claimed subject matter. Instead, the claimed subject matter should be understood by reference to appropriate portions of the Specification and drawings, as well as to each claim.

In some embodiments, a system includes a plurality of photovoltaic modules installed on a roof deck, each of the plurality of photovoltaic modules comprising at least one solar cell, and a backsheet, wherein the at least one solar cell is above the backsheet, and wherein the backsheet includes a headlap portion, wherein the headlap portion comprises a first longitudinal groove and a second longitudinal groove, wherein the first longitudinal groove is spaced apart from the second longitudinal groove to define a nail zone between the first longitudinal groove and the second longitudinal groove, and wherein each of the photovoltaic modules is configured to receive at least one fastener within the nail zone for installing the photovoltaic module to the roof deck.

In some embodiments, a depth of at least one of the first longitudinal groove or the second longitudinal groove is 0.1 mm to 0.2 mm. In some embodiments, a width of the nail zone between the first longitudinal groove and the second longitudinal groove is 3 cm to 6 cm. In some embodiments, each of the first longitudinal groove and the second longitudinal groove extends from one end of the backsheet to an opposite end of the backsheet.

In some embodiments, the backsheet is composed of thermoplastic polyolefin (TPO). In some embodiments, the headlap portion an integral portion of the backsheet. In some embodiments, the backsheet includes a base portion, and

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wherein the headlap portion is attached to the base portion. In some embodiments, the at least one solar cell and the base portion are laminated, and wherein the headlap portion is not laminated. In some embodiments, the headlap portion does not include a pigment.

In some embodiments, a photovoltaic module includes at least one solar cell, and a backsheet, wherein the at least one solar cell is above the backsheet, wherein the backsheet comprises a headlap portion, wherein the headlap portion comprises a first longitudinal groove and a second longitudinal groove, wherein the first longitudinal groove is spaced apart from the second longitudinal groove on the backsheet to define a nail zone between the first longitudinal groove and the second longitudinal groove, and wherein the nail zone is configured to receive at least one fastener that penetrates the photovoltaic module to secure the photovoltaic module to a roof deck.

In some embodiments, the backsheet comprises thermoplastic polyolefin (TPO). In some embodiments, a depth of at least one of the first longitudinal groove or the second longitudinal groove is 0.1 mm to 0.2 mm. In some embodiments, a width of the nail zone between the first longitudinal groove and the second longitudinal groove is 3 cm to 6 cm. In some embodiments, each of the first longitudinal groove and the second longitudinal groove extends from one end of the backsheet to an opposite end of the backsheet. In some embodiments, the headlap portion an integral portion of the backsheet. In some embodiments, the backsheet includes a base portion, and wherein the headlap portion is attached to the base portion. In some embodiments, the at least one solar cell and the base portion are laminated, and wherein the headlap portion is not laminated. In some embodiments, the headlap portion does not include a pigment.

In some embodiments, a method comprises obtaining a first sheet; forming a first longitudinal groove in the first sheet; forming a second longitudinal groove in the first sheet, the first longitudinal groove spaced apart from the second longitudinal groove to define a first nail zone between the first longitudinal groove and the second longitudinal groove; forming a third longitudinal groove in the first sheet; forming a fourth longitudinal groove in the first sheet, the third longitudinal groove spaced apart from the fourth longitudinal groove to define a second nail zone between the third longitudinal groove and the fourth longitudinal groove, wherein the first nail zone is spaced apart from the second nail zone; cutting the sheet to form a first backsheet and a second backsheet, wherein the first backsheet includes the first nail zone, and wherein the second backsheet includes the second nail zone; and laminating a first solar cell with the first backsheet to form a first photovoltaic module, wherein the first photovoltaic module includes a headlap portion, and wherein the first nail zone is on the headlap portion.

In some embodiments, the method further includes laminating a second solar cell with the second backsheet to form a second photovoltaic module, wherein the second photovoltaic module includes a headlap portion, and where the second nail zone is on the headlap portion of the second photovoltaic module.

In some embodiments, the forming the first longitudinal groove and the forming the second longitudinal groove occur simultaneously. In some embodiments, the forming the first longitudinal groove occurs before the forming the second longitudinal groove.

In some embodiments, the forming the third longitudinal groove and the forming the fourth longitudinal groove occur

simultaneously. In some embodiments, the forming the third longitudinal groove occurs before the forming the fourth longitudinal groove.

In some embodiments, the forming the first longitudinal groove, the forming the second longitudinal groove, the forming of the third longitudinal groove, and the forming of the fourth longitudinal groove occur simultaneously. In some embodiments, the forming the first longitudinal groove, the forming the second longitudinal groove, the forming of the third longitudinal groove, and/or the forming of the fourth longitudinal groove occurs at different periods.

BRIEF DESCRIPTION OF THE FIGURES

This section refers to the drawings that form a part of this disclosure, and which illustrate some of the embodiments of structure, materials, and/or methods of the present invention described herein.

FIG. 1 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 2 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 3 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 4 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 5 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 6 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 7 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 8 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 9 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 10 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 11 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 12 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 13 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 14 is a schematic top view of a roofing module, in accordance with some embodiments of the invention.

FIG. 15 is a schematic top view of a roofing system including a plurality of roofing modules, in accordance with some embodiments of the invention.

FIG. 16 is a photograph of a backsheet material, in accordance with embodiments of the invention.

FIG. 17 is a photograph of the backsheet material of FIG. 16, over which a top layer has been laminated, in accordance with some embodiments of the invention.

FIG. 18 is a photograph illustrating an embodiment of a method of forming a groove in a photovoltaic module.

FIG. 19 illustrates an embodiment of a method of making a roofing module.

FIGS. 20A and 20B are photographs of some embodiments of backsheets including nail zones for roofing module.

FIG. 21 illustrates an embodiment of a method of making a photovoltaic module.

FIG. 22 is a photograph of some embodiments of a photovoltaic module.

FIG. 23 are photographs of some embodiments of backsheets including nail zones for roofing modules.

DETAILED DESCRIPTION

In addition to the benefits and improvements that the Specification discloses, other objects and advantages that the Specification provides will become apparent from the following description taken in conjunction with the accompanying figures. Although the description discloses and describes detailed embodiments of the present disclosure, the disclosed embodiments are merely illustrative of the disclosure that may be embodied in various forms. In addition, each of the examples given regarding the various embodiments of the disclosure is intended to be illustrative, and not restrictive.

As used herein, a “steep slope” roof or roof deck is a roof or roof deck having a pitch of Y/X, where Y and X are in a ratio of 4:12 to 20:12, where Y corresponds to the “rise” of the roof or roof deck, and where X corresponds to the “run” of the roof or roof deck.

As used herein, a “sloped” roof or roof deck is a roof or roof deck that is not flat, but has a pitch that is less than the pitch of a steep slope roof or roof deck.

As used herein, a “flat” roof or roof deck is a roof or roof deck that is substantially flat.

In some embodiments, the present invention provides a roofing module comprising a pair of grooves defining a nail zone therebetween.

In some embodiments, the grooves comprise a pair of longitudinal grooves. In some embodiments, the longitudinal grooves are spaced apart from one another, thereby defining the nail zone.

In some embodiments, the roofing module comprises a photovoltaic module. In some embodiments, the roofing module is a photovoltaic module shown and described in either or both of U.S. application Ser. No. 17/831,307, filed Jun. 2, 2022, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2022-0393637 on Dec. 8, 2022; and/or U.S. application Ser. No. 18/169,718, filed Feb. 15, 2023, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2023-0203815 on Jun. 29, 2023, the disclosures of each of which are incorporated by reference herein in their entireties. In some embodiments, the photovoltaic module includes a structure, composition, components, and/or function similar to those of one or more embodiments of the photovoltaic modules and shingles disclosed in PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, titled “Building Integrated Photovoltaic System,” owned by GAF Energy LLC; U.S. Pat. No. 11,251,744 to Bunea et al., issued Feb. 15, 2022, titled “Photovoltaic Shingles and Methods of Installing Same, owned by GAF Energy LLC;” and U.S. Pat. No. 11,870,227, issued Jan. 9, 2024, entitled “Building Integrated Photovoltaic System,” owned by GAF Energy LLC, the disclosures of each of which are incorporated by reference herein in their entireties.

In some embodiments, the roofing module is a non-photovoltaic module. In some embodiments, the non-photovoltaic module is a cuttable roofing module. In some embodiments, the roofing module includes a cap layer and a core layer. In some embodiments, the cuttable roofing module, the cap layer, and/or the core layer, are as shown and described in U.S. application Ser. No. 17/831,307, filed Jun. 2, 2022, titled “Roofing Module System,” and pub-

lished under U.S. Patent Application Publication No. 2022-0393637 on Dec. 8, 2022, owned by GAF Energy LLC; U.S. application Ser. No. 18/169,718, filed Feb. 15, 2023, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2023-0203815 on Jun. 29, 2023, owned by GAF Energy LLC; and/or U.S. application Ser. No. 18/352,894, filed Jul. 14, 2023, titled “Solar Roofing System with Fiber Composite Roofing Shingles,” owned by GAF Energy LLC, the disclosures of each of which are incorporated by reference herein in their entireties.

In some embodiments, the photovoltaic module comprises at least one solar cell. In some embodiments, the at least one solar cell is electrically active—that is, the solar cell and/or associated electronics convert light, such as sunlight, into electricity by the photovoltaic effect. In some embodiments, the at least one solar cell is electrically inactive—that is, the solar cell and/or associated electronics do not convert light into electricity.

In some embodiments, one or more of the solar cells of the photovoltaic module are electrically active. In some embodiments, all of the solar cells of the photovoltaic module are electrically active. In some embodiments, one or more of the solar cells of the photovoltaic module are electrically inactive. In some embodiments, all of the solar cells of the photovoltaic module are electrically inactive. In some embodiments, at least one of the solar cells of the photovoltaic module is active while at least one of the solar cells of the photovoltaic module is electrically inactive.

In some of the embodiments, the solar cells of the roofing module are positioned above a backsheet. In some embodiments, the solar cells are laminated to the backsheet. In some embodiments, the backsheet comprises one or more of ethylene propylene diene monomer (EPDM) rubber, polymethyl methacrylate (PMMA), polyethylene terephthalate (PET), ethylene tetrafluoroethylene (ETFE), thermoplastic polyolefin (TPO), and/or combinations of two or more thereof.

In some embodiments, the backsheet is a single layer. In some embodiments, the backsheet is multiple layers.

In some embodiments, the groove or grooves are formed on a cap layer. In some embodiments, the groove or grooves are formed on the cap layer, which is on a core layer.

In some embodiments, the roofing module is configured to be installed above a roof deck with one or more fasteners. In some embodiments, the fasteners comprise one or more of nails, screws, staples, rivets, and/or combinations thereof.

In some embodiments, the backsheet comprises a headlap portion. In some embodiments, the headlap portion defines the nail zone. In some embodiments, the headlap portion defines the nail zone between the longitudinal grooves that are formed on the backsheet.

In some embodiments, the headlap portion comprises a first longitudinal groove, and a second longitudinal groove spaced apart from the first longitudinal groove on the backsheet, thereby to define the nail zone between the first longitudinal groove and the second longitudinal groove. In some embodiments, the nail zone is configured to receive at least one fastener that penetrates the backsheet of the photovoltaic module when the photovoltaic module is installed above the roof deck in a roofing system, as well as the roof deck, thereby to secure the photovoltaic module to the roof deck.

In some embodiments, a depth of at least one of the first longitudinal groove and/or the second longitudinal groove is 0.01 mm. In some embodiments, the depth is 0.05 mm. In some embodiments, the depth is 0.1 mm. In some embodiments, the depth is 0.2 mm. In some embodiments, the depth

is 0.3 mm. In some embodiments, the depth is 0.4 mm. In some embodiments, the depth is 0.5 mm. In some embodiments, the depth is 0.6 mm. In some embodiments, the depth is 0.7 mm. In some embodiments, the depth is 0.8 mm. In some embodiments, the depth is 0.9 mm. In some embodiments, the depth is 1.0 mm. In some embodiments, the depth is 1.1 mm. In some embodiments, the depth is 1.2 mm. In some embodiments, the depth is 1.3 mm. In some embodiments, the depth is 1.4 mm. In some embodiments, the depth is 1.5 mm. In some embodiments, the depth is 1.6 mm. In some embodiments, the depth is 1.7 mm. In some embodiments, the depth is 1.8 mm. In some embodiments, the depth is 1.9 mm. In some embodiments, the depth is 2.0 mm. In some embodiments, the depth is 3.0 mm. In some embodiments, the depth is 4.0 mm. In some embodiments, the depth is 5.0 mm.

In some embodiments, the depth is more than 0.01 mm. In some embodiments, the depth is more than 0.05 mm. In some embodiments, the depth is more than 0.1 mm. In some embodiments, the depth is more than 0.2 mm. In some embodiments, the depth is more than 0.3 mm. In some embodiments, the depth is more than 0.4 mm. In some embodiments, the depth is more than 0.5 mm. In some embodiments, the depth is more than 0.6 mm. In some embodiments, the depth is more than 0.7 mm. In some embodiments, the depth is more than 0.8 mm. In some embodiments, the depth is more than 0.9 mm. In some embodiments, the depth is more than 1.0 mm. In some embodiments, the depth is more than 1.1 mm. In some embodiments, the depth is more than 1.2 mm. In some embodiments, the depth is more than 1.3 mm. In some embodiments, the depth is more than 1.4 mm. In some embodiments, the depth is more than 1.5 mm. In some embodiments, the depth is more than 1.6 mm. In some embodiments, the depth is more than 1.7 mm. In some embodiments, the depth is more than 1.8 mm. In some embodiments, the depth is more than 1.9 mm. In some embodiments, the depth is more than 2.0 mm. In some embodiments, the depth is more than 3.0 mm. In some embodiments, the depth is more than 4.0 mm. In some embodiments, the depth is more than 5.0 mm.

In some embodiments, the depth is less than 0.01 mm. In some embodiments, the depth is less than 0.05 mm. In some embodiments, the depth is less than 0.1 mm. In some embodiments, the depth is less than 0.2 mm. In some embodiments, the depth is less than 0.3 mm. In some embodiments, the depth is less than 0.4 mm. In some embodiments, the depth is less than 0.5 mm. In some embodiments, the depth is less than 0.6 mm. In some embodiments, the depth is less than 0.7 mm. In some embodiments, the depth is less than 0.8 mm. In some embodiments, the depth is less than 0.9 mm. In some embodiments, the depth is less than 1.0 mm. In some embodiments, the depth is less than 1.1 mm. In some embodiments, the depth is less than 1.2 mm. In some embodiments, the depth is less than 1.3 mm. In some embodiments, the depth is less than 1.4 mm. In some embodiments, the depth is less than 1.5 mm. In some embodiments, the depth is less than 1.6 mm. In some embodiments, the depth is less than 1.7 mm. In some embodiments, the depth is less than 1.8 mm. In some embodiments, the depth is less than 1.9 mm. In some embodiments, the depth is less than 2.0 mm. In some embodiments, the depth is less than 3.0 mm. In some embodiments, the depth is less than 4.0 mm. In some embodiments, the depth is less than 5.0 mm.

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In some embodiments, the depth is 1% to 95% of the thickness of the layer or layers in which the groove is made (e.g., a backsheet). In some embodiments, the depth is 5% to 95% of the thickness. In some embodiments, the depth is 10% to 95% of the thickness. In some embodiments, the

In some embodiments, the width is 0.5 inch to 1.7 inches. In some embodiments, the width is 0.6 inch to 1.7 inches. In some embodiments, the width is 0.7 inch to 1.7 inches. In some embodiments, the width is 0.8 inch to 1.7 inches. In some embodiments, the width is 0.9 inch to 1.7 inches. In some embodiments, the width is 1.0 inch to 1.7 inches. In some embodiments, the width is 1.1 inches to 1.7 inches. In some embodiments, the width is 1.2 inches to 1.7 inches. In some embodiments, the width is 1.3 inches to 1.7 inches. In

In some embodiments, the width is 0.5 inch to 0.8 inch. In some embodiments, the width is 0.6 inch to 0.8 inch. In some embodiments, the width is 0.7 inch to 0.8 inch.

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sheet is an uppermost layer of the photovoltaic module when the photovoltaic module is installed on or above a roof deck in a roofing system.

In some embodiments, the frontsheet is a single layer. In some embodiments, the frontsheet is multiple layers.

In some embodiments, there is an adhesive between the frontsheet and the encapsulant.

In some embodiments, a roofing system includes a roof deck, and at least one roofing module installed on or above the roof deck. In some embodiments, the roof deck is a steep slope roof deck. In some embodiments, the roof deck is a sloped roof deck. In some embodiments, the roof deck is a flat roof deck.

In some embodiments, the at least one roofing module comprises one roofing module. In some embodiments, the at least one roofing module comprises two roofing modules. In some embodiments, the at least one roofing module comprises three roofing modules. In some embodiments, the at least one roofing module comprises four roofing modules. In some embodiments, the at least one roofing module comprises five roofing modules. In some embodiments, the at least one roofing module comprises more than five roofing modules. In some embodiments, at least one of the roofing modules is a photovoltaic module. In some embodiments, at least one of the roofing modules is a non-photovoltaic module. In some embodiments, at least one of the roofing modules is a photovoltaic module, and at least one of the roofing modules is a non-photovoltaic module.

In some embodiments, when the roofing modules are installed on or above the roof deck in the roofing system, at least some of the roofing modules are installed next to one another in a same row on the roof deck. In some embodiments, when the roofing modules are installed on or above the roof deck, at least some of the roofing modules are installed in multiple, different rows, where one row is above the other row on the roof deck. In some embodiments, when the roofing modules are installed on or above the roof deck in multiple rows, at least some of the roofing modules in one row overlap at least some of the roofing modules in another row. In some embodiments, when the roofing modules are installed on or above the roof deck in multiple rows, at least some of the roofing modules in one row overlap headlap portions of at least some of the roofing modules in another row (e.g., such that a part of one roofing module covers at least a part of the headlap portion of another roofing module).

In some embodiments, one or more fasteners are installed through the nail zone of one or more of the roofing modules, thereby to fasten the roofing module to the roof deck. In some embodiments, the one or more fasteners penetrate the nail zone. In some embodiments, the one or more fasteners penetrate the roof deck. In some embodiments, the one or more fasteners penetrate an underlayment between the roof deck and the roofing module (e.g., when the roofing module is installed above the roof deck). In some embodiments, an underlayment is not installed or otherwise disposed between the roof deck and the roofing module.

In some embodiments, the present invention provides a method. In some embodiments, the method comprises obtaining a backsheet material. In some embodiments, the method comprises forming the first groove in the headlap portion of the backsheet material (e.g., in a portion of the backsheet material that will be the headlap when the backsheet material is used to form a photovoltaic module). In some embodiments, the method comprises forming the second groove in the headlap portion of the backsheet material. In some embodiments, the grooves are formed

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simultaneously (e.g., the forming of the first groove occurs at a same time as the forming of the second groove). In some embodiments, one groove is formed prior to the other groove.

In some embodiments, either or both of the grooves are formed in the backsheet material to which no other layers are added. In some embodiments, either or both of the grooves are formed in the backsheet material to which one or more other layers are laminated before formation of the backsheet.

In some embodiments, at least one groove is formed by a tool. In some embodiments, the tool is an awl. In some embodiments, the tool is a bit. In some embodiments, the tool is a blade. In some embodiments, the tool is a cutter. In some embodiments, the tool is knife. In some embodiments, the tool is a blade. In some embodiments, the tool is a razor. In some embodiments, the tool includes a sharp tip. In some embodiments, the tool is disposed in a fixture. In some embodiments, the fixture is disposed within a manufacturing line. In some embodiments, the manufacturing line is a manufacturing line for a backsheet material. In some embodiments, the backsheet material is cut into backsheets. In some embodiments, the backsheet material is cut into one or more backsheets in a same manufacturing line. In some embodiments, the backsheet material is cut into one or more backsheets in a different manufacturing line.

In some embodiments, after forming one or both of the grooves, the grooved backsheet material is cut, thereby to form a grooved backsheet. In some embodiments, the frontsheet, encapsulant and/or the at least one solar cell are laminated with the backsheet, thereby to form the photovoltaic module. In some embodiments, before the backsheet material is cut, one or more layers are laminated to the grooved backsheet material.

With reference to the figures, FIG. 1 shows a schematic view of a roofing module, according to some embodiments of the invention. As shown in the figure, in some embodiments, the roofing module **100** comprises a photovoltaic module. In some embodiments, the roofing module **100** comprises at least one solar cell **200**, and a backsheet **300**. In some embodiments, the solar cell **200** is positioned on the backsheet **300**. In some embodiments, the solar cell **200** is laminated to the backsheet **300**. In some embodiments, the roofing module **100** is a non-photovoltaic module. In some embodiments, the roofing module **100** includes a cap layer over a core layer.

As shown in FIG. 1, in some embodiments, the backsheet **300** includes a headlap portion **310**. In some embodiments, the headlap portion **310** includes two longitudinal grooves **320** (e.g., a first groove and a second groove) formed thereon. As shown in the figure, in some embodiments, the longitudinal grooves **320** are spaced apart from one another, thereby to define a nail zone **330** between the longitudinal grooves **320**. In some embodiments, the nail zone **330** is configured to receive at least one fastener that penetrates the roofing module **100**, thereby to secure or fasten the roofing module **100** to a roof deck.

As shown in FIGS. 1 and 2, in some embodiments, the longitudinal grooves **320** extend between a first end **340** of the backsheet **300** and a second end **350** of the backsheet **300**, the first end **340** spaced apart from and opposite to the second end **350** along a major dimension of the shingle, such that the longitudinal grooves **320** extend an entire length of the roofing module **100** (e.g., the longitudinal grooves extend an entire distance between the first end **340** and the second end **350**). In some embodiments, as shown in FIG. 1, each of the longitudinal grooves **320** is a single, continuous line segment. However, as shown in FIG. 2, either or both

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of the longitudinal grooves 320 may be two or more line segments (e.g., a discontinuous line).

In some embodiments, as shown in FIGS. 3-14, one or both of the longitudinal grooves 320 do not extend from end 340 to end 350 of the backsheet 300 (e.g., one or both of the longitudinal grooves 320 do not extend an entire length of the roofing module). In some embodiments, either or both of the longitudinal grooves 320 may be two or more lines segments (e.g., a discontinuous line as shown in FIG. 2).

Referring to FIG. 3, in some embodiments, one of the grooves 320 extends from the first end 340 to a location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the first end 340 to the second end 350.

Referring to FIG. 4, in some embodiments, one of the grooves 320 extends from the first end 340 to a location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the second end 350 to a location between the first end 340 and the second end 350.

Referring to FIG. 5, in some embodiments, one of the grooves 320 extends from the first end 340 to a location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the first end 340 to a location between the first end 340 and the second end 350.

Referring to FIG. 6, in some embodiments, one of the grooves 320 extends from the first end 340 to a location between the first end 340 and the second end 350, while the other of the grooves 320 extends from a first location between the first end 340 and the second end 350 to a second location between the first end 340 and the second end 350.

Referring to FIG. 7, in some embodiments, one of the grooves 320 extends from the second end 350 to a location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the first end 340 to the second end 350.

Referring to FIG. 8, in some embodiments, one of the grooves 320 extends from the second end 350 to a location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the second end 350 to a location between the first end 340 and the second end 350.

Referring to FIG. 9, in some embodiments, one of the grooves 320 extends from the second end 350 to a location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the first end 340 to a location between the first end 340 and the second end 350.

Referring to FIG. 10, in some embodiments, one of the grooves 320 extends from the second end 350 to a location between the first end 340 and the second end 350, while the other of the grooves 320 extends from a first location between the first end 340 and the second end 350 to a second location between the first end 340 and the second end 350.

Referring to FIG. 11, in some embodiments, one of the grooves 320 extends from a first location between the first end 340 and the second end 350 to a second location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the first end 340 to the second end 350.

Referring to FIG. 12, in some embodiments, one of the grooves 320 extends from a first location between the first end 340 and the second end 350 to a second location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the second end 350 to a location between the first end 340 and the second end 350.

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Referring to FIG. 13, in some embodiments, one of the grooves 320 extends from a first location between the first end 340 and the second end 350 to a second location between the first end 340 and the second end 350, while the other of the grooves 320 extends from the first end 340 to a location between the first end 340 and the second end 350.

Referring to FIG. 14, in some embodiments, one of the grooves 320 extends from a first location between the first end 340 and the second end 350 to a second location between the first end 340 and the second end 350, while the other of the grooves 320 extends from a third location between the first end 340 and the second end 350 to a fourth location between the first end 340 and the second end 350.

FIG. 15 is a schematic view of a roofing system including a plurality of roofing modules, installed above a roof deck, in accordance with some embodiments of the invention. As shown in the figure, the roofing system 400 includes a plurality of roofing modules 100 installed above a roof deck 500. As shown, one or more fasteners 502 penetrate the nail zones 330 of the roofing modules 100, and penetrate the roof deck 500, thereby to install the roofing modules 100 to the roof deck 500. In some embodiments, the roofing system 400 includes an underlayment 504 installed between the roofing modules and the roof deck 500. In some embodiments, the roofing system 400 does not include an underlayment installed between the roofing modules and the roof deck 500.

FIG. 16 is a photograph of a backsheet material including two longitudinal grooves—a grooved backsheet material—in accordance with embodiments of the invention. FIG. 17 is a photograph of the grooved backsheet material over which a top layer has been laminated, in accordance with some embodiments of the invention. As shown, the longitudinal grooves, and thus the nail zone therebetween, remains visible before and after lamination, such that the nail zone will be visible in a roofing module in accordance with the above-described embodiments of the invention. In some embodiments, the term “visible” means visible to a human eye without the aid of magnifying equipment.

FIG. 18 is a photograph of some embodiments of a method of forming one or more grooves 620 in a surface 622 of a roofing shingle 600 to form a nail zone. In some embodiments, the surface 622 is an upper surface of the roofing shingle 600. In some embodiments, the grooves 620 are formed by one or more tools 630. In some embodiments, the tool 630 includes a tip 632. In some embodiments, the tip 632 is a sharp tip. In some embodiments, the tip 632 is a pointed tip. In some embodiments, the tool 630 is a knife, a razor, or cutting tip. In some embodiments, the tool 630 creates the groove 620 by cutting, digging, or scratching.

Referring to FIG. 19, in some embodiments, a method of forming one or more roofing shingles 700 with nail zones is illustrated. In some embodiments, the roofing shingle 700 is a photovoltaic shingle or solar shingle. In some embodiments, the roofing shingle 700 is a non-photovoltaic shingle. In some embodiments, a first sheet 702 is provided. In some embodiments, the first sheet 702 is composed of a polymer. In some embodiments, the first sheet 702 is composed of one or more of ethylene propylene diene monomer (EPDM) rubber, polymethyl methacrylate (PMMA), polyethylene terephthalate (PET), ethylene tetrafluoroethylene (ETFE), thermoplastic polyolefin (TPO), and/or combinations of two or more thereof. In some embodiments, the first sheet 702 includes an upper surface 704. In some embodiments, one or more sets of grooves 706 are formed in the upper surface 704 of the first sheet 702. In some embodiments, the grooves 706 are formed by one or more of the instruments and/or

methods as described herein. In some embodiments, each set of grooves **706** includes a pair of the grooves **706**. In some embodiments, each pair of grooves **706** forms a nail zone **708**. In some embodiments, the first sheet **702** is cut into dimensions at one or more cut lines **703** to form a plurality of backsheets **710**. In some embodiments, the first sheet **702** is cut into desired dimensions such that one portion of each of the backsheets **710** is configured to receive other layers of the roofing shingle **700**, such as a layer of one or more solar cells **712**, and another portion of each of the backsheets **710** is a headlap portion **714**. In some embodiments, the headlap portion **714** includes the nail zone **708** formed by the grooves **706**. In some embodiments, the grooves **706** are visible in the headlap portion **714** to a user, such as a roofing installer.

In some embodiments, the backsheet **710** undergoes lamination with one or more other layers to fabricate the roofing shingle **700**. In some embodiments, lamination of the roofing shingle **700** occurs at 165° C. In some embodiments, lamination of the roofing shingle **700** occurs from 150° C. to 180° C. In some embodiments, lamination of the roofing shingle **700** occurs at a pressure of 14 psi. In some embodiments, lamination of the roofing shingle **700** occurs at a pressure of 10 psi to 20 psi. In some embodiments, lamination of the roofing shingle **700** occurs for 20 minutes. In some embodiments, lamination of the roofing shingle **700** occurs for 15 minutes to 25 minutes. In some embodiments, the grooves **706** do not disappear or diminish during lamination of the roofing shingle **700**. In some embodiments, material of the backsheet **710** or other layers of the roofing shingle **700** do not fill in or substantially fill in the grooves **706** during lamination of the roofing shingle **700** so as to diminish, substantially diminish, or eliminate their visibility.

FIGS. **20A** and **20B** illustrate embodiments of backsheets **710** composed of TPO formulations with different base resins that were tested and evaluated to determine which include nail zone grooves that are visible after lamination. In some embodiments, referring to Table 1 below and FIG. **20A**, the results indicate that a melting point within 2° C. of a lamination temperature of 165° C., coupled with a high flexural modulus of 480 MPa for P7 TPO with a base resin of Q402F resulted in visible grooves. In some embodiments, referring to Table 1 below and FIG. **20B**, the results indicate that a melting point within 2° C. of the lamination temperature of 165° C., coupled with a lower flexural modulus of 150 MPa for P26 TPO with a base resin of Adflex Q200F resulted in visible grooves, but with lower visibility than that of the embodiment of FIG. **20A**.

TABLE 1

TPO Formulation	Base Resin	Resin melt point (° C.)	Resin Flexural Modulus (MPa)	Result
P26	Adflex Q200F	163	150	Visibility of Grooves
P7	Q402F	163	480	Visibility of Grooves

In some embodiments, a method includes:
 obtaining a first sheet;
 forming a first longitudinal groove in the first sheet;
 forming a second longitudinal groove in the first sheet, the first longitudinal groove spaced apart from the second

longitudinal groove to define a first nail zone between the first longitudinal groove and the second longitudinal groove;

forming a third longitudinal groove in the first sheet;
 forming a fourth longitudinal groove in the first sheet, the third longitudinal groove spaced apart from the fourth longitudinal groove to define a second nail zone between the third longitudinal groove and the fourth longitudinal groove, wherein the first nail zone is spaced apart from the second nail zone;
 cutting the sheet to form a first backsheet and a second backsheet, wherein the first backsheet includes the first nail zone, and wherein the second backsheet includes the second nail zone;
 laminating a first solar cell with the first backsheet to form a first photovoltaic module, wherein the first photovoltaic module includes a headlap portion, and wherein the first nail zone is on the headlap portion; and
 in some embodiments, laminating a second solar cell with the second backsheet to form a second photovoltaic module, wherein the second photovoltaic module includes a headlap portion, and where the second nail zone is on the headlap portion of the second photovoltaic module.

Referring to FIGS. **21** through **23**, in some embodiments, a method of forming one or more roofing shingles **800** with nail zones is illustrated. In some embodiments, the roofing shingle **800** is a photovoltaic shingle or solar shingle. In some embodiments, the roofing shingle **800** is a non-photovoltaic shingle. In some embodiments, a first sheet **802** and a second sheet **803** are provided. In some embodiments, the first sheet **802** is composed of a polymer. In some embodiments, the first sheet **802** is composed of one or more of ethylene propylene diene monomer (EPDM) rubber, polymethyl methacrylate (PMMA), polyethylene terephthalate (PET), ethylene tetrafluoroethylene (ETFE), thermoplastic polyolefin (TPO), and/or combinations of two or more thereof.

In some embodiments, the second sheet **803** is composed of a polymer. In some embodiments, the second sheet **803** is composed of one or more of ethylene propylene diene monomer (EPDM) rubber, polymethyl methacrylate (PMMA), polyethylene terephthalate (PET), ethylene tetrafluoroethylene (ETFE), thermoplastic polyolefin (TPO), and/or combinations of two or more thereof.

Referring to FIG. **23**, in some embodiments, the second sheet **803** is composed of TPO with a P3 formulation and Hifax™ CA10A base resin. In some embodiments, the second sheet **803** is composed of TPO with a P26 formulation and an Adflex™ Q200F base resin. In some embodiments, the second sheet **803** is composed of TPO with a P7 formulation and Adflex™ Q402F base resin. In some embodiments, the second sheet **803** is composed of TPO with a P22 formulation and ProFax™ base resin. In some embodiments, the second sheet **803** is composed of TPO with a C12 formulation and Hifax™ CA10A base resin. In some embodiments, the second sheet **803** includes a dark color. In some embodiments, the second sheet **803** includes a black color. In some embodiments, the second sheet **803** includes a color that enables grooves **814** formed therein to be visible.

In some embodiments, the first sheet **802** and the second sheet **803** are separate from one another. In some embodiments, the first sheet **802** is cut to desired dimensions at one or more cut lines **801** to produce one or more base portions **804** that are configured to receive other layers of the roofing shingle **800**, such as a layer of one or more solar cells **806**

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in a reveal portion, and side lap portions **808**, **810**. In some embodiments, the base portion **804** and the solar cells **806** are laminated.

In some embodiments, the second sheet **803** includes an upper surface **812**. In some embodiments, one or more sets of grooves **814** are formed in the upper surface **812** of the second sheet **803**. In some embodiments, the grooves **814** are formed by one or more of the instruments and methods as described herein. In some embodiments, each set of grooves **814** includes a pair of the grooves **814**. In some embodiments, each pair of grooves **814** forms a nail zone **816**. In some embodiments, the second sheet **803** is cut into dimensions at one or more cut lines **801** to form a plurality of headlap portions **818**. In some embodiments, the headlap portion **818** is not laminated. In some embodiments, the grooves **814** of the nail zone **816** are not subjected to lamination.

In some embodiments, the headlap portion **818** is attached to the base portion **804**. In some embodiments, a lower edge **820** of the headlap portion **818** is attached to an upper edge **822** of the base portion **804**. In some embodiments, the headlap portion **818** is attached to the base portion **804** by heat pressure. In some embodiments, the headlap portion **818** is attached to the base portion **804** by welding. In some embodiments, the headlap portion **818** is attached to the base portion **804** by ultrasonic welding. In some embodiments, the headlap portion **818** is attached to the base portion **804** by an adhesive. In some embodiments, the headlap portion **818** does not include a UV stabilizer. In some embodiments, the headlap portion **818** does not include a pigment. In some embodiments, the base portion **804** includes a pigment. In some embodiments, the base portion **804** includes a UV stabilizer.

In some embodiments, a method includes:

- obtaining a first sheet;
- forming a first longitudinal groove in the first sheet;
- forming a second longitudinal groove in the first sheet, the first longitudinal groove spaced apart from the second longitudinal groove to define a first nail zone between the first longitudinal groove and the second longitudinal groove;
- forming a third longitudinal groove in the first sheet;
- forming a fourth longitudinal groove in the first sheet, the third longitudinal groove spaced apart from the fourth longitudinal groove to define a second nail zone between the third longitudinal groove and the fourth longitudinal groove, wherein the first nail zone is spaced apart from the second nail zone;
- cutting the first sheet to form a first headlap portion and a second headlap portion, wherein the first headlap portion includes the first nail zone, and wherein the second headlap portion includes the second nail zone;
- obtaining a second sheet;
- laminating a first solar cell with the second sheet;
- attaching the first headlap portion to the second sheet to form a first photovoltaic module;
- in some embodiments, obtaining a third sheet;
- laminating a second solar cell with the third sheet; and
- attaching the second headlap portion to the third sheet to form a second photovoltaic module.

Variations, modifications, and alterations to embodiments of the present disclosure described above will make themselves apparent to those skilled in the art. All such variations, modifications, alterations, and the like are intended to fall within the spirit and scope of the present disclosure, limited solely by the appended claims.

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While several embodiments of the present disclosure have been described, it is understood that these embodiments are illustrative only, and not restrictive, and that many modifications may become apparent to those of ordinary skill in the art. For example, all dimensions discussed herein are provided as examples only, and are intended to be illustrative and not restrictive.

Any feature or element that is positively identified in this description may also be specifically excluded as a feature or element of an embodiment of the present as defined in the claims.

The disclosure described herein may be practiced in the absence of any element or elements, limitation or limitations, which is not specifically disclosed herein. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the disclosure.

What is claimed is:

1. A system, comprising: a plurality of photovoltaic modules installed on a roof deck, each of the plurality of photovoltaic modules comprising: at least one solar cell, and a backsheet,

wherein the at least one solar cell is above the backsheet, and

wherein the backsheet includes a headlap portion, wherein the headlap portion comprises a first surface, and a first longitudinal groove in the first surface and a second longitudinal groove in the first surface,

wherein the first longitudinal groove is spaced apart from the second longitudinal groove to define a nail zone between the first longitudinal groove and the second longitudinal groove, and

wherein each of the photovoltaic modules is configured to receive at least one fastener within the nail zone for installing the photovoltaic module to the roof deck, and wherein each of the plurality of photovoltaic modules does not include granules on at least the first surface of the headlap portion.

2. The system of claim 1, wherein a depth of at least one of the first longitudinal groove or the second longitudinal groove is 0.1 mm to 0.2 mm.

3. The system of claim 1, wherein a width of the nail zone between the first longitudinal groove and the second longitudinal groove is 3 cm to 6 cm.

4. The system of claim 1, wherein each of the first longitudinal groove and the second longitudinal groove extends from one end of the backsheet to an opposite end of the backsheet.

5. The system of claim 1, wherein the backsheet is composed of thermoplastic polyolefin (TPO).

6. The system of claim 1, wherein the headlap portion an integral portion of the backsheet.

7. The system of claim 1, wherein the backsheet includes a base portion, and wherein the headlap portion is attached to the base portion.

8. The system of claim 7, wherein the at least one solar cell and the base portion are laminated, and wherein the headlap portion is not laminated.

9. The system of claim 8, wherein the headlap portion does not include a pigment.

10. A photovoltaic module, comprising: at least one solar cell, and

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- a backsheet, wherein the at least one solar cell is above the backsheet,
 wherein the backsheet comprises a headlap portion,
 wherein the headlap portion comprises a first surface,
 and a first longitudinal groove in the first surface and a
 second longitudinal groove in the first surface,
 wherein the first longitudinal groove is spaced apart from
 the second longitudinal groove on the backsheet to
 define a nail zone between the first longitudinal groove
 and the second longitudinal groove, and
 wherein the nail zone is configured to receive at least one
 fastener that penetrates the photovoltaic module to
 secure the photovoltaic module to a roof deck, and
 wherein each of the plurality of photovoltaic modules
 does not include granules on at least the first surface of
 the headlap portion.
11. The photovoltaic module of claim 10, wherein the
 backsheet comprises thermoplastic polyolefin (TPO).
12. The photovoltaic module of claim 10, wherein a depth
 of at least one of the first longitudinal groove or the second
 longitudinal groove is 0.1 mm to 0.2 mm.
13. The photovoltaic module of claim 10, wherein a width
 of the nail zone between the first longitudinal groove and the
 second longitudinal groove is 3 cm to 6 cm.
14. The photovoltaic module of claim 10, wherein each of
 the first longitudinal groove and the second longitudinal
 groove extends from one end of the backsheet to an opposite
 end of the backsheet.
15. The photovoltaic module of claim 10, wherein the
 headlap portion an integral portion of the backsheet.
16. The photovoltaic module of claim 10, wherein the
 backsheet includes a base portion, and wherein the headlap
 portion is attached to the base portion.
17. The photovoltaic module of claim 16, wherein the at
 least one solar cell and the base portion are laminated, and
 wherein the headlap portion is not laminated.
18. The photovoltaic module of claim 16, wherein the
 headlap portion does not include a pigment.

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19. A method, comprising:
 obtaining a first sheet wherein the first sheet includes a
 first surface;
 forming a first longitudinal groove in the first surface of
 the first sheet;
 forming a second longitudinal groove in the first surface
 of the first sheet, the first longitudinal groove spaced
 apart from the second longitudinal groove to define a
 first nail zone between the first longitudinal groove and
 the second longitudinal groove; forming a third longi-
 tudinal groove in the first surface of the first sheet;
 forming a fourth longitudinal groove in the first surface
 of the first sheet, the third longitudinal groove spaced
 apart from the fourth longitudinal groove to define a
 second nail zone between the third longitudinal groove
 and the fourth longitudinal groove,
 wherein the first nail zone is spaced apart from the second
 nail zone;
 cutting the first sheet to form a first backsheet and a
 second backsheet, wherein the first backsheet includes
 the first nail zone, and wherein the second backsheet
 includes the second nail zone; and
 laminating a first solar cell with the first backsheet to form
 a first photovoltaic module,
 wherein the first photovoltaic module includes a headlap
 portion, and
 wherein the first nail zone is on the headlap portion, and
 wherein each of the plurality of photovoltaic modules
 does not include granules on at least the first surface of
 the headlap portion.
20. The method of claim 19, further comprising laminat-
 ing a second solar cell with the second backsheet to form a
 second photovoltaic module, wherein the second photovol-
 taic module includes a headlap portion, and where the
 second nail zone is on the headlap portion of the second
 photovoltaic module.

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